Chemistry of the Earth: Lecture 6

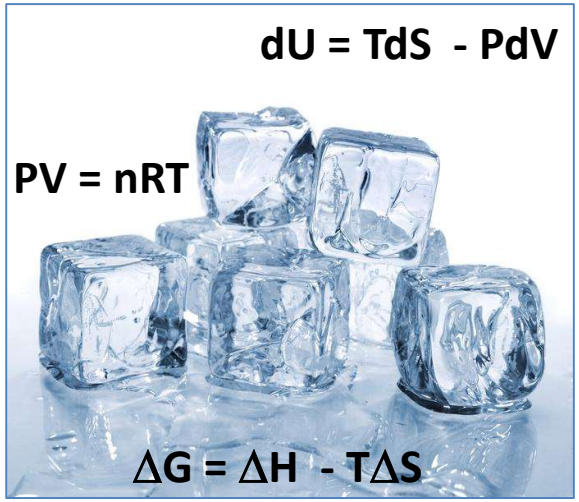
Chemical Equilibrium (Thermodynamics)

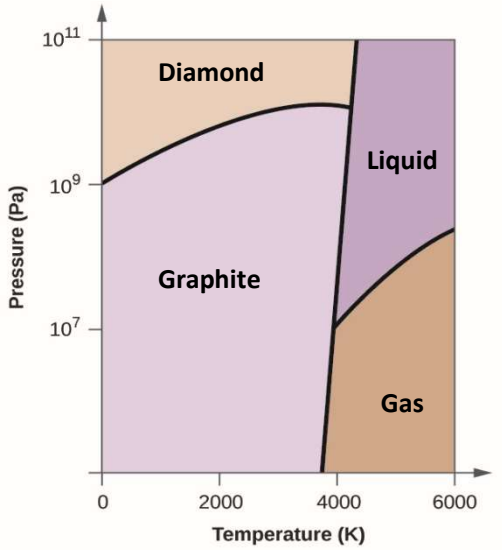


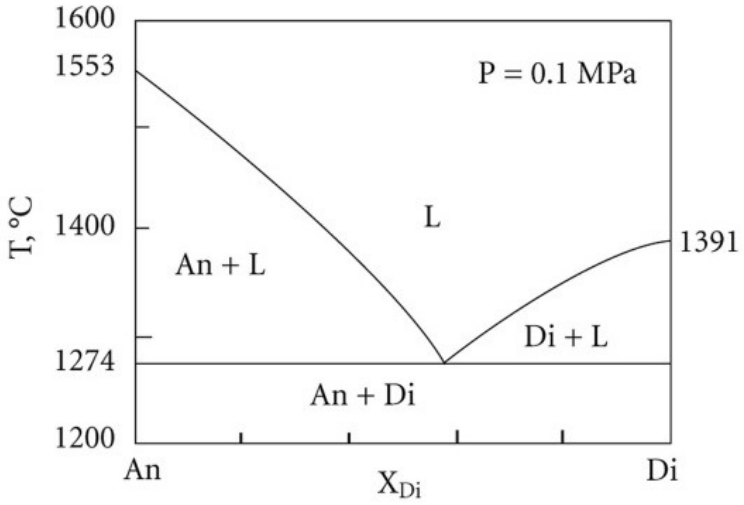
Part 1: Introduction to Thermodynamics

OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Understand what is meant by equilibrium in a chemical reaction
- A basic introduction to thermodynamics
- Understand how equilibrium is represented in a phase diagram
- Understand how geochemical systems respond to changes in temperature and pressure (mineral polymorphs, melting)







READING AND RESOURCES

Grotzinger & Jordan (Basic: Minimum reading)	**p.60-61 (polymorphs), p.97-98 (rock melting)**	
Gill (Recommended)	**Chapter 2, p. 14-30**	
Misra	Chapter 4	(all are a lot more detailed than the level expected in this class)
White	Chapter 2	
McSween & Richardson (Recommended)	Chapters 3 and 9	

Internet resources

"Teaching Phase Equilibria"
http://serc.carleton.edu/research_education/equilibria/index.html

EQUILIBRIUM – WHY WE CARE

Equilibrium – a state of matter that does not change with time

We want to be able to predict the outcome of a chemical reaction, hence the state of matter at some point in the Earth given defined temperature, pressure and other conditions:

- Predict at what temperature a rock will melt
- Predict the sequence of minerals that will crystallize from a melt to form an igneous rock
- Predict the new minerals that will form when a rock undergoes metamorphism
- Predict the way a rock will dissolve during weathering, the compositions of waters, and which minerals precipitate from waters

THERMODYNAMICS

Thermo – heat Dynamics – movement

The study of energy (heat) and energy changes. Thermodynamics is used to describe how systems respond to changes in their environment.

ENERGY and **HEAT**: Heat is thermal energy.

- Heat results from random motions of molecules and atoms within a substance.
- Atoms also have chemical energy stored in chemical bonds.

Unit of energy: **JOULE**

Joule: Energy transferred to an object when a force of one newton acts on that object in the direction of its motion through a distance of one metre.
Newton: Force needed to accelerate 1 kg of mass at a rate of 1 m s^{-2} in the direction of the applied force.

$$1\ J = 1\ kg\ m^2\ s^{-2} = 1\ N\ m$$



James Prescott Joule (1818 –1889), physicist and brewer, born in Salford

Wikipedia

SYSTEM: The part of the universe that we are considering

Examples: A lava lake, a granite pluton, a single mineral grain, a sedimentary deposit, the ocean, the Earth's core, the atmosphere, the Earth-Moon system.

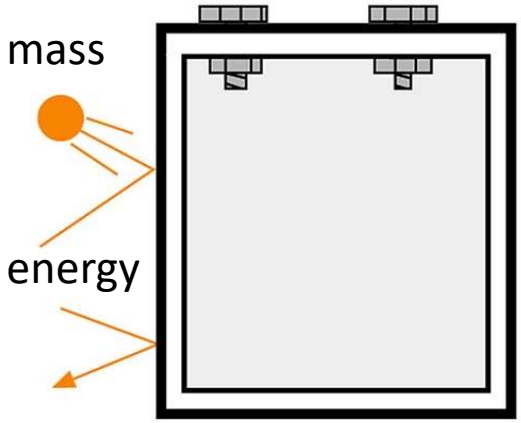
ISOLATED, CLOSED AND OPEN SYSTEMS:

Relationship of a system to its environment. A system may be able to exchange energy (heat) and / or matter with its surroundings.

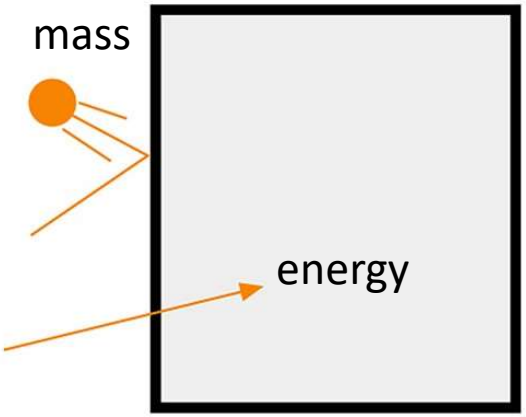
<u>Isolated System</u>
No exchange of energy or matter

<u>Closed System</u>
Can exchange energy but not matter

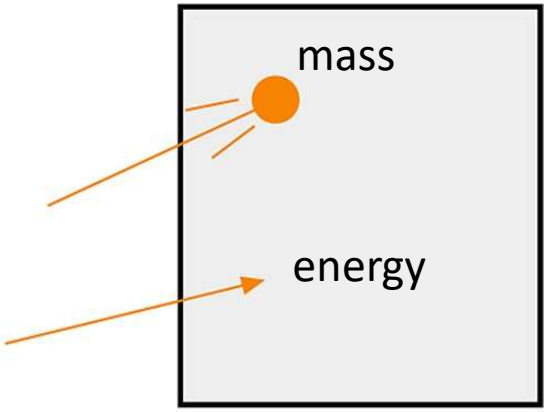
<u>Open System</u>
Can exchange energy and matter



Isolated System



Closed System



Open System

Geochemistry, First Edition. William M. White.
© 2013 John Wiley & Sons, Ltd. Published 2013 by John Wiley & Sons, Ltd.

VARIABLE PROPERTIES OF A SYSTEM:

There are two different categories of variable properties:

Extensive variables:
Depend on the size of the system

- Mass
- Volume
- Energy

Intensive variables:
Do not depend on the size of the system

- Temperature
- Pressure
- Viscosity
- Density (= Mass/Volume)

An example of the relationship between these variables is the **Ideal Gas Law**:

$$\mathbf{PV = nRT}$$

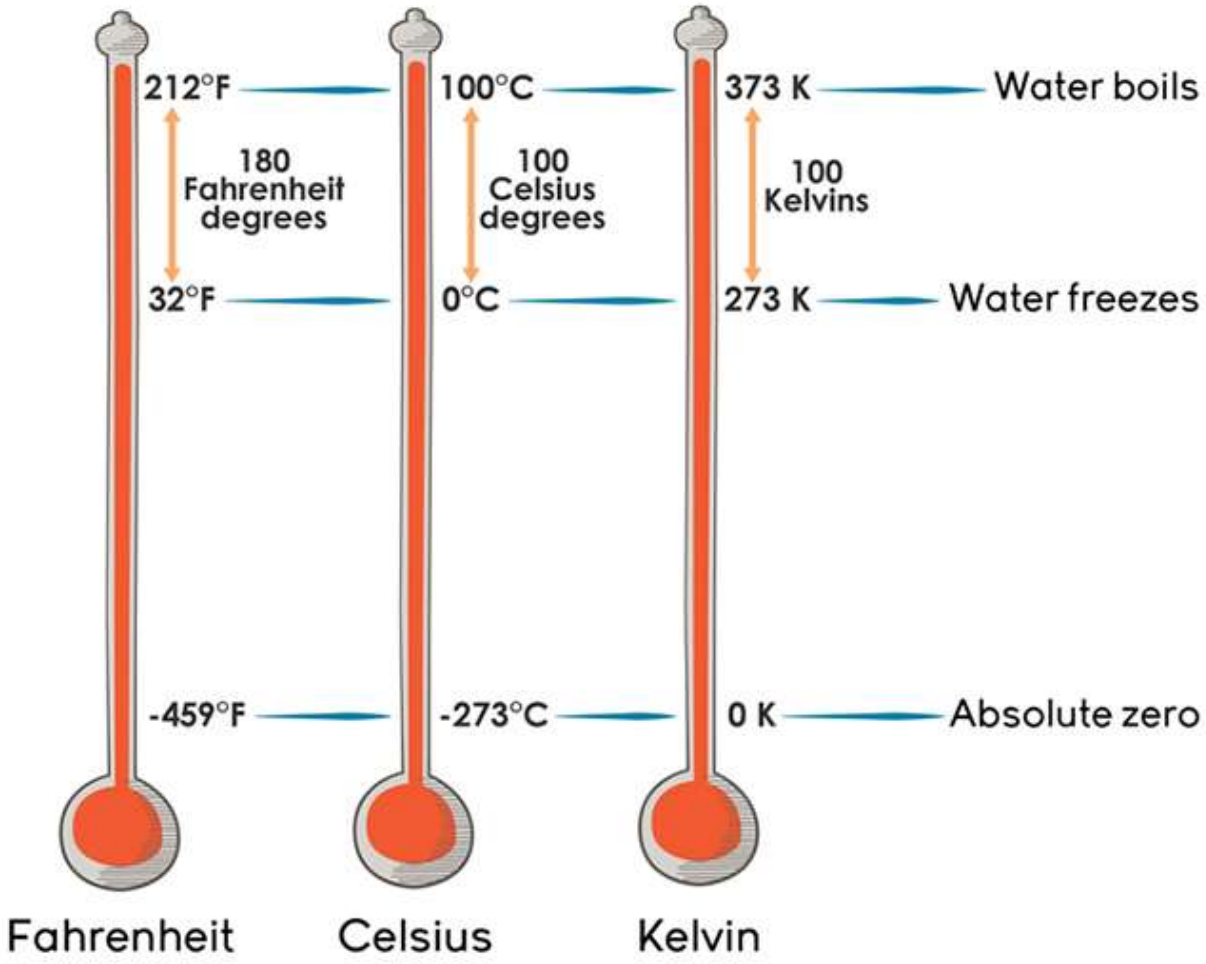
P = pressure
V = volume
n = number of moles
R = Gas Constant (8.314 J mol^{-1})
T = temperature in Kelvin

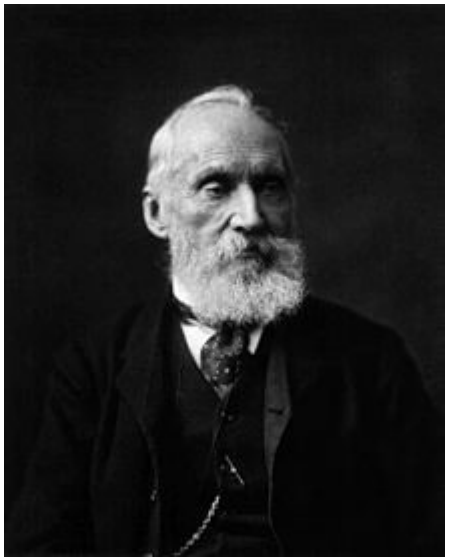


USGS: Mount St. Helens May 19, 1982

Temperature in thermodynamics – the Kelvin scale and Absolute Zero

Based on Ideal Gas Law – relationship between T and V of an ideal gas. V = 0 at T = 0.





William Thomson, Lord Kelvin, 1824-1907

Wikipedia

www.learner.org/courses/chemistry

THE LAWS OF THERMODYNAMICS

FIRST LAW OF THERMODYNAMICS

Principle of the **conservation of energy** – energy is neither created nor destroyed in chemical reactions and during physical changes.

$$dU = dQ - PdV$$

FIRST LAW of THERMODYNAMICS

Internal energy, U: for a mineral this is the sum of potential energy stored in bonds and kinetic energy of atomic vibrations.

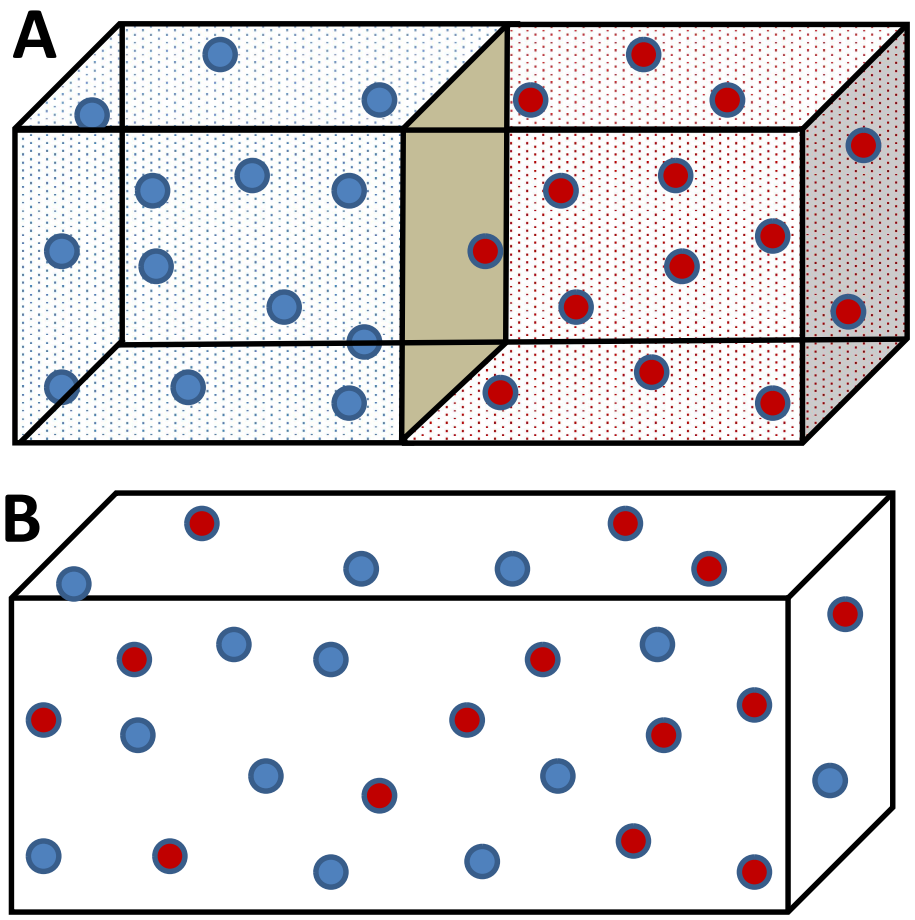
A small change in internal energy of the system (dU) is equal to the heat flowing into the system from the surroundings (Q) plus the mechanical work done on the system (W): $dU = dQ + dW$

The most important form of work in geologic systems comes from changes in pressure (P) and volume (V): $dW = -PdV$.

(Note that McSween uses E instead of U)

SECOND LAW OF THERMODYNAMICS

Concept of **ENTROPY** (symbol **S**)



Consider two boxes of gas separated by a partition (A).
When the partition is removed (B), the gases mix.
But the reverse process is not possible.

There is no way to account for this behaviour in the First Law. We need a function to describe this sense of direction.

The direction of spontaneous, irreversible, change is described by the term **entropy**, given the symbol **S**.
Entropy is a measure of *how the energy is distributed* in a system.

SECOND LAW OF THERMODYNAMICS

ENTROPY (S) Units of S: J mol^{-1} K^{-1}

- Nature favours spontaneous change.
- The direction of change is towards a higher entropy of the system and its surroundings.
- Entropy can be thought of as a measure of randomness or disorder in a system.

$$dS \geq dQ/T$$

SECOND LAW of THERMODYNAMICS

In the example above, transport of the gases occurred because this *resulted in an increase in entropy*.

An example of how entropy changes with different states of matter (physical change):

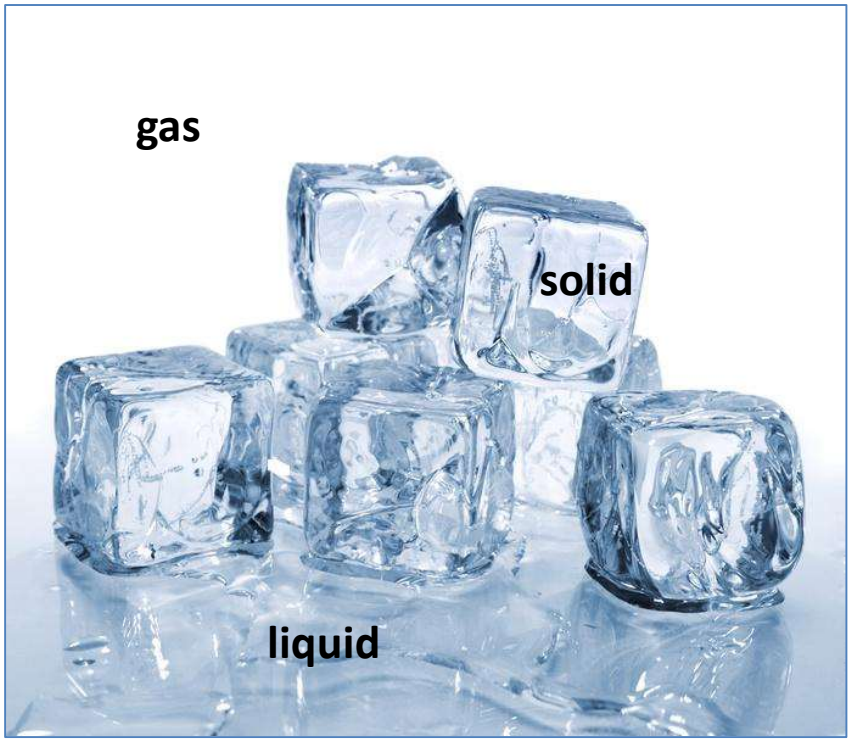
Molar entropy (entropy per mole) of H_2O:

$S\ (H_2O, \text{steam}) = 188.8\ J\ mol^{-1}\ K^{-1}$

$S\ (H_2O, \text{water}) = 70.0\ J\ mol^{-1}\ K^{-1}$

$S\ (H_2O, \text{ice}) = 44.7\ J\ mol^{-1}\ K^{-1}$

Entropy relates to the amount of motion of water molecules in each state (solid / liquid / gas)



ice-cube-melting.wikispaces.com

COMBINING THE FIRST AND SECOND LAWS

First Law: $dU = dQ - PdV$

Second Law: $dS \geq dQ/T$

Combined: $dU \leq TdS - PdV$

Since most geological process occur very slowly and small changes are close to being reversible, we can assume that:

$$\mathbf{dU = TdS - PdV}$$

FUNDAMENTAL EQUATION of THERMODYNAMICS

U – internal energy
T – temperature in Kelvin
S – entropy
P – pressure
V – volume

SOME OTHER IMPORTANT CONCEPTS IN THERMODYNAMICS

ENTHALPY (H): The total heat content of a substance. Units: Joules (J)

$$H = U + PV$$

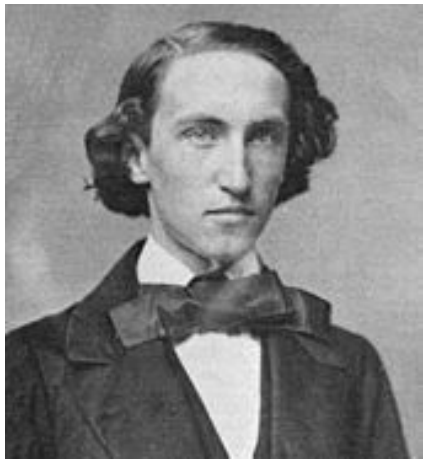
If a change takes place at constant pressure, the change in enthalpy is equivalent to the heat gained or lost by the system, and

$$dH = dQ$$

GIBBS FREE ENERGY: The amount of energy available for a system to do useful work at constant T and P.

$$G = H - TS$$

Gibbs free energy defines the conditions for equilibrium (reversible) and spontaneous (irreversible) change.
Hence, G defines in which direction a spontaneous change will occur.



Josiah Willard Gibbs
(1839-1903) Wikipedia

GIBBS FREE ENERGY: The amount of energy available for a system to do useful work at constant T and P.

$$G = H - TS$$

For a change in the system,

$$\boxed{\Delta G = \Delta H - T\Delta S}$$

When $\Delta G = 0$, the system is in **chemical equilibrium**.

When $\Delta G < 0$ (i.e. when ΔG is negative), there is energy available to produce a spontaneous change.
A chemical reaction will proceed in the direction of lower Gibbs free energy.

Another useful relationship in thermodynamics:

$$\boxed{dG = VdP - SdT}$$

THERMODYNAMICS IN A NUTSHELL

$PV = nRT$ — **Ideal Gas Law**

$dU = dQ - PdV$ — **First Law of Thermodynamics**

$dS \geq dQ/T$ — **Second Law of Thermodynamics**

$dU = TdS - PdV$

$dG = VdP - SdT$

$\Delta G = \Delta H - T\Delta S$

d = a very small change
Δ = a change

P = pressure
V = volume
n = number of moles
R = Gas Constant (8.314 J mol^{-1})
T = temperature **in Kelvin**
U = internal energy (sometimes written as E)
Q = heat flow from surroundings
S = entropy
H = enthalpy
G = Gibbs Free Energy

Chemistry of the Earth: Lecture 6

Chemical Equilibrium (Thermodynamics)

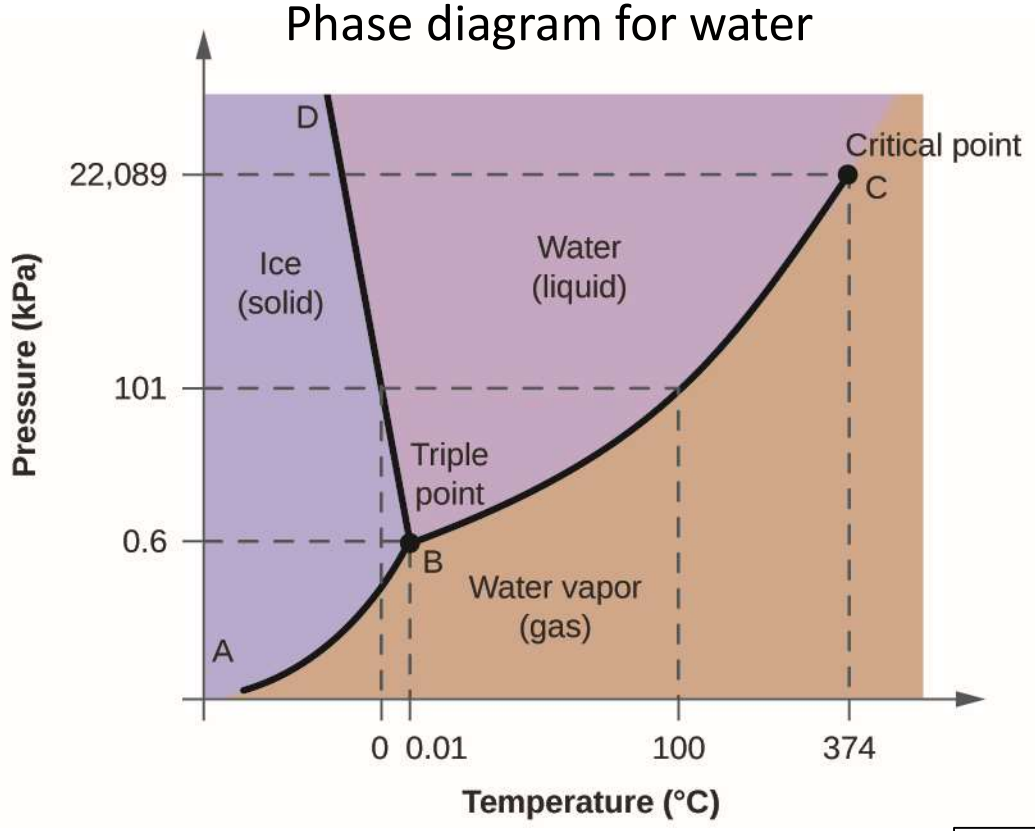


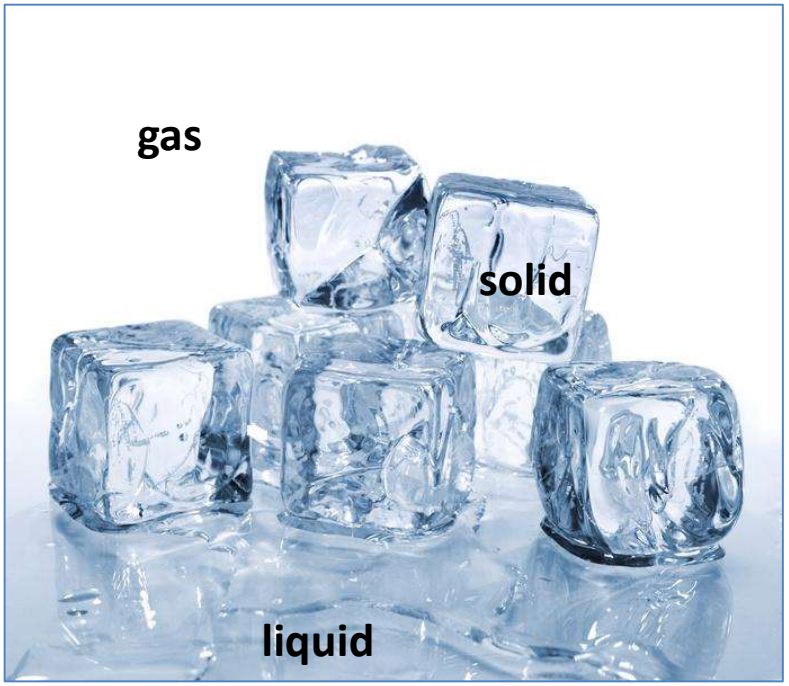
Part 2: Introduction to Phase Diagrams

PHASE DIAGRAMS

PHASE: A phase is a physically distinct body of matter that is (at least in principle) physically separable from the rest of the system.

Phase diagram for water





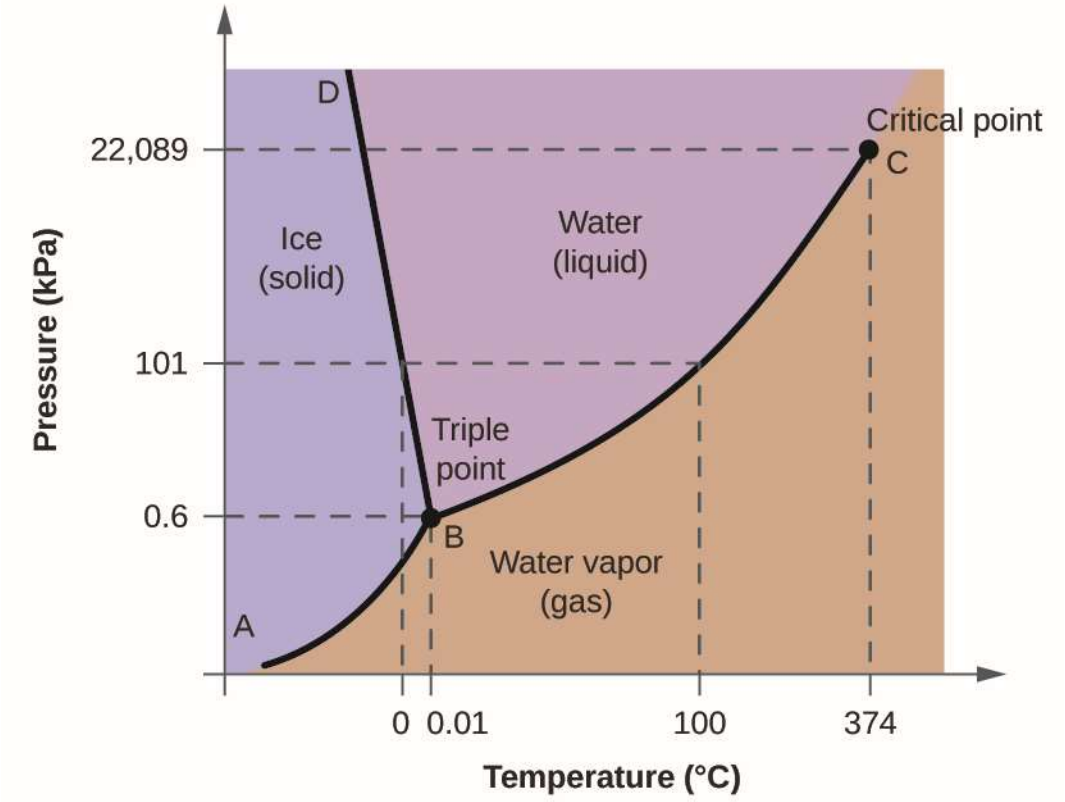
ice-cube-melting.wikispaces.com

1 Pa = 10^{-5} bar
1 kPa = 10^{-2} bar
100 kPa = 1 bar

PHASE DIAGRAMS:

Representation of the regions of stability of one or more phases

- Shows the areas where each phase is thermodynamically stable – the equilibrium state of the system at a given P and T
- Lines show where reactions and/or physical changes occur



Example: Phase diagram for H_2O

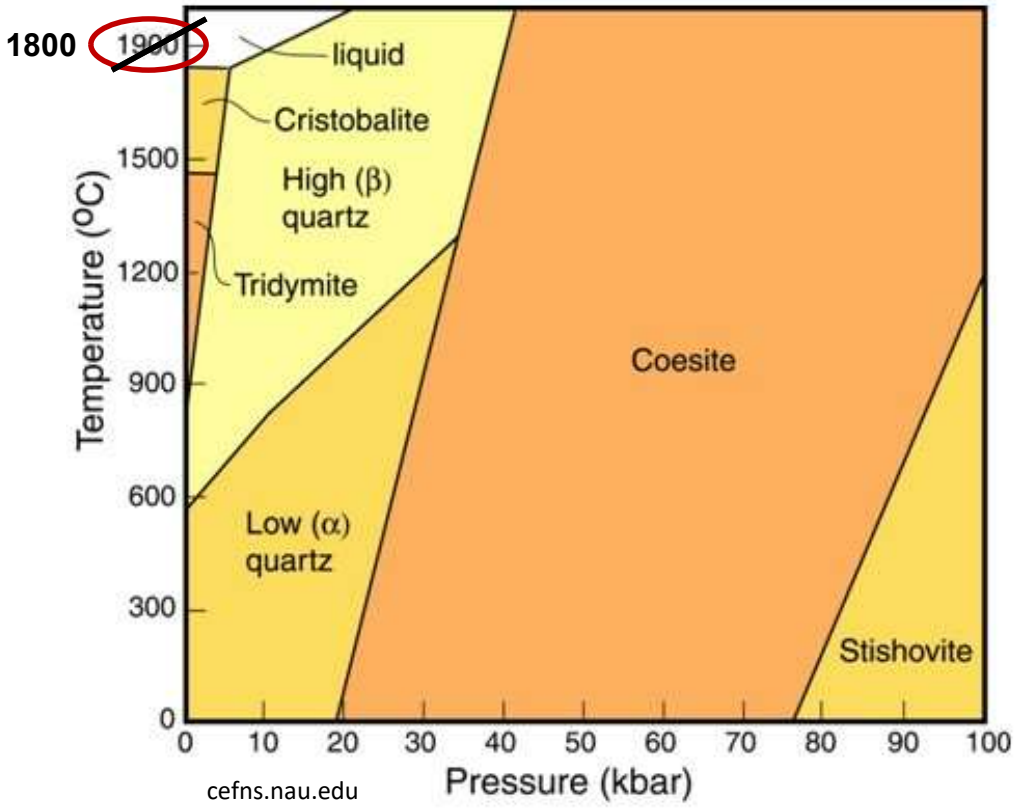
Shows the temperatures and pressures at which H_2O is stable as ice, water and steam.

Curves show where two phases are in equilibrium e.g. melting / freezing and boiling / condensation curves.

Triple point – all three phases are in equilibrium.

PHASE DIAGRAMS FOR MINERALS AND ROCKS

SiO_2 system



(note that the number 1900 on the y axis should be 1800)

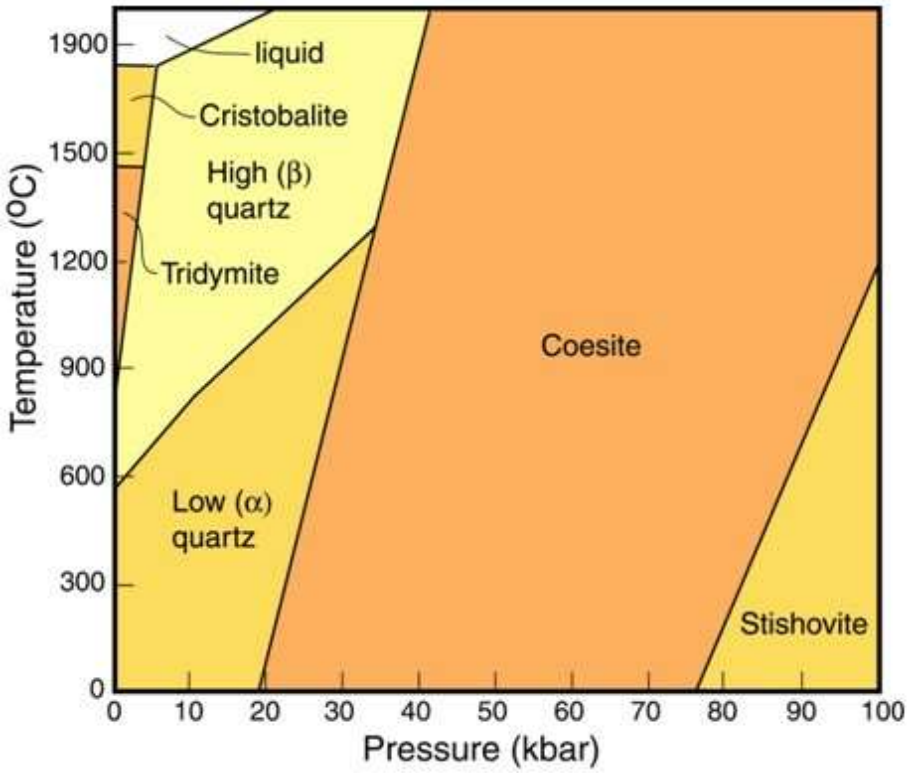
- The diagram shows temperature vs pressure (T vs P)
- The composition throughout the diagram is SiO_2 (silica)
- The diagram shows the range of temperatures and pressures where the different silica minerals (silica **polymorphs**) are stable
- In this case, each silica mineral is a different **phase**
- The diagram also shows the conditions where SiO_2 melts

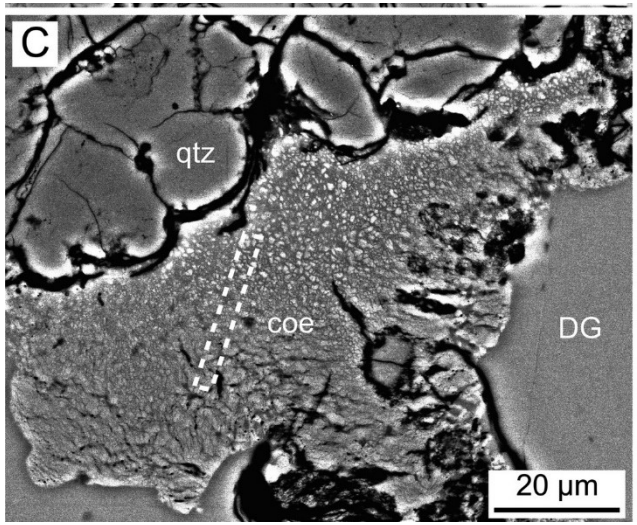
Pressure is given here in kbar. One kbar = 1000 bars, or 1000 atmospheres.
1 kbar is about 3-4 km depth in the Earth's crust.
Pressure is also commonly given in Mega-Pascals (MPa) or Giga-Pascals (GPa). 1 GPa = 10 kbar.

Barringer Meteorite Crater, Arizona:

First natural discovery of coesite and stishovite



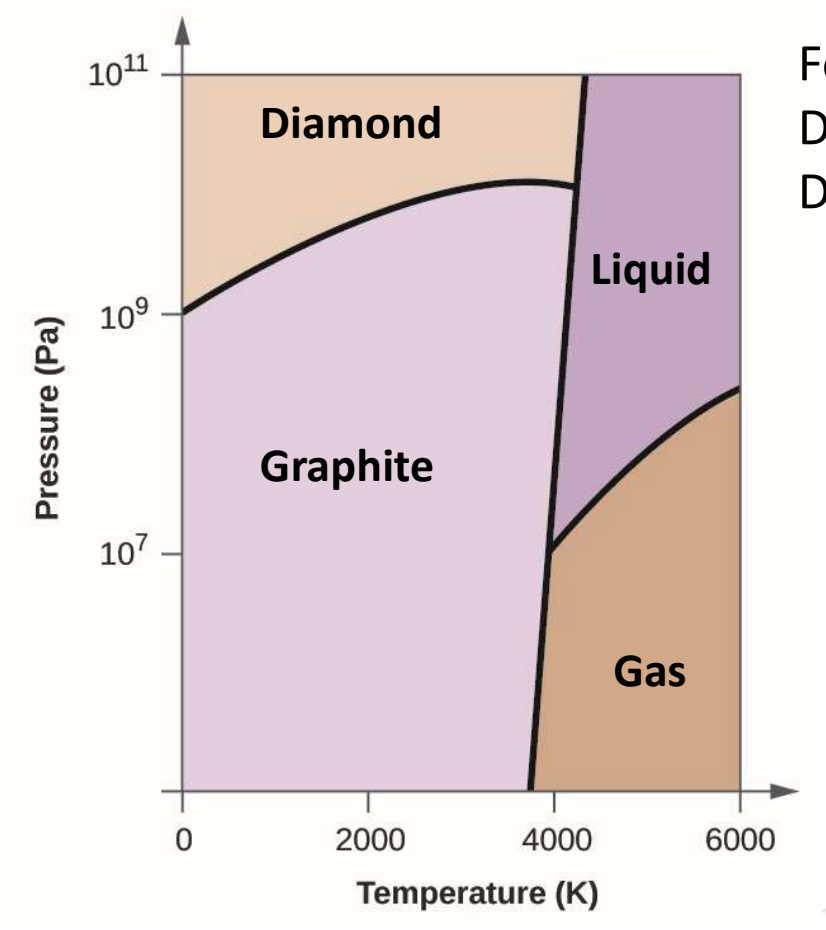




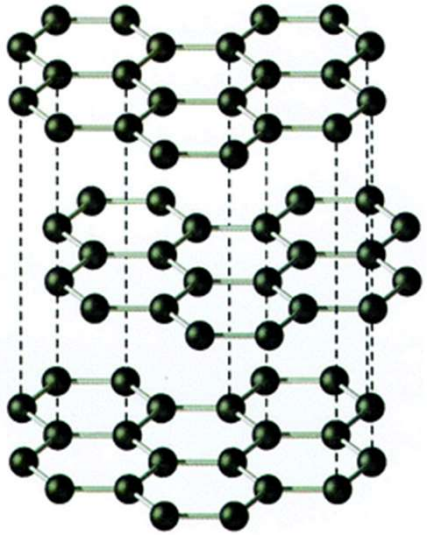
Coesite in suevite from the Ries impact structure (Germany)

Fazio et al. (2017) Meteoritics & Planetary Science, 52, 1437-1448

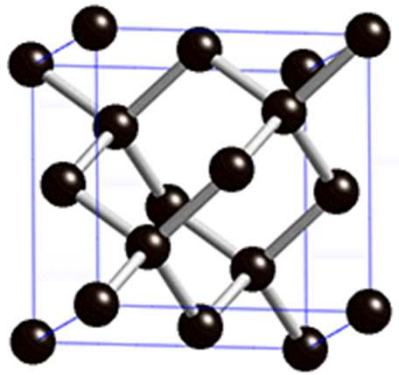
PHASE DIAGRAM FOR CARBON



Four phases are shown:
Diamond, graphite, liquid and vapour.
Diamond and graphite are carbon *polymorphs*.



graphite



diamond

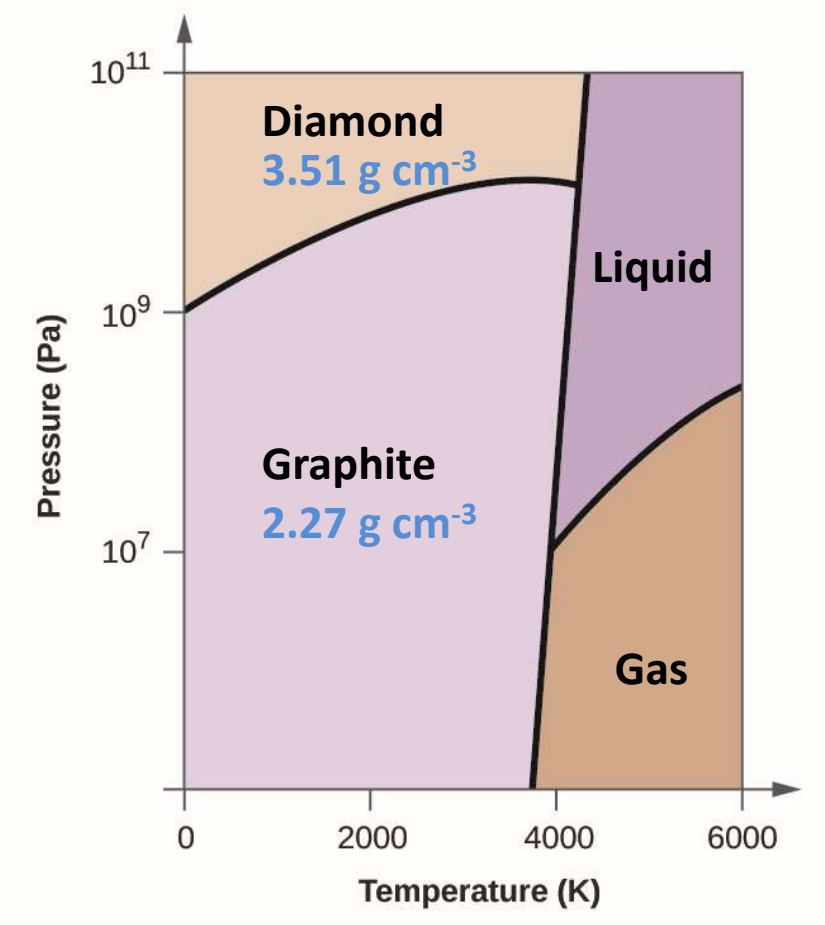


Wikipedia



eBay

PHASE DIAGRAM FOR CARBON



- Graphite is the stable solid phase at low pressures
- Graphite transforms to diamond at high pressures
- Graphite has a much lower density than diamond. Higher pressures favour the denser phase.

- Diamond is not thermodynamically stable at the conditions at the Earth's surface.
- The presence of diamond at the earth's surface is because the *rate* at which diamond transforms to graphite is so slow that it does not happen (kinetics)

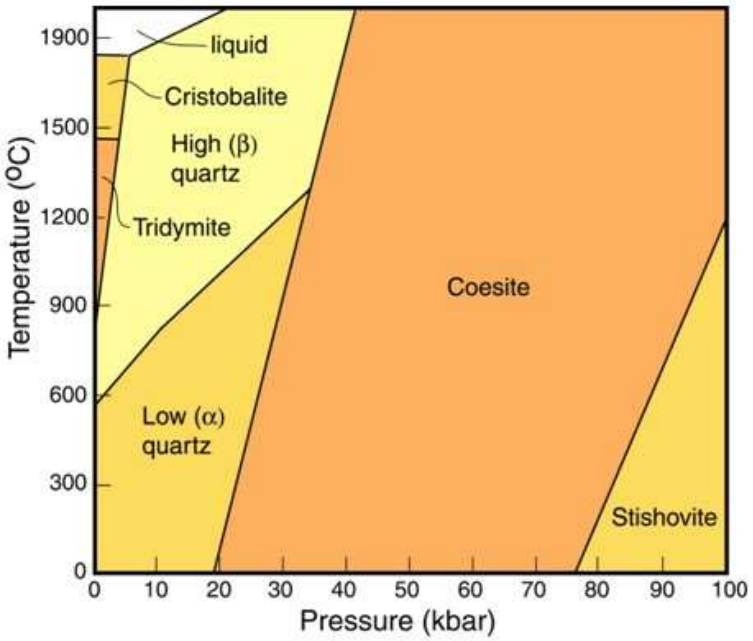


"Diamond - South Africa - Finsch Mine" by StrangerThanKindness

COMPONENTS:

The number of **components** in a system is the minimum number of chemical species required to describe the composition of all the phases in the system.

Phase diagram for SiO_2

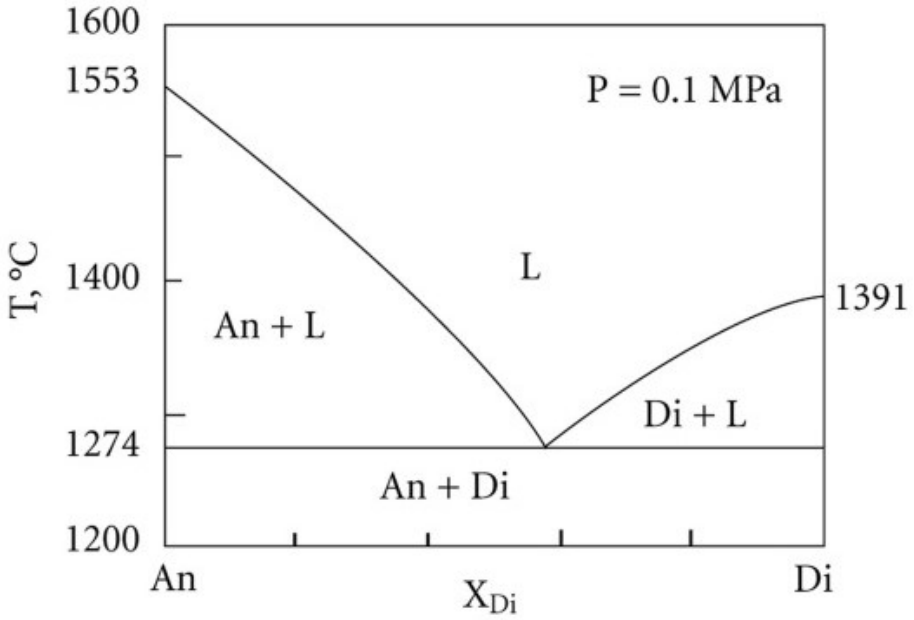


One component – SiO_2

Diagram shows T vs P

Composition (X) is constant throughout the diagram (X = SiO_2)

Phase diagram for Anorthite-Diopside



Two components – An and Di

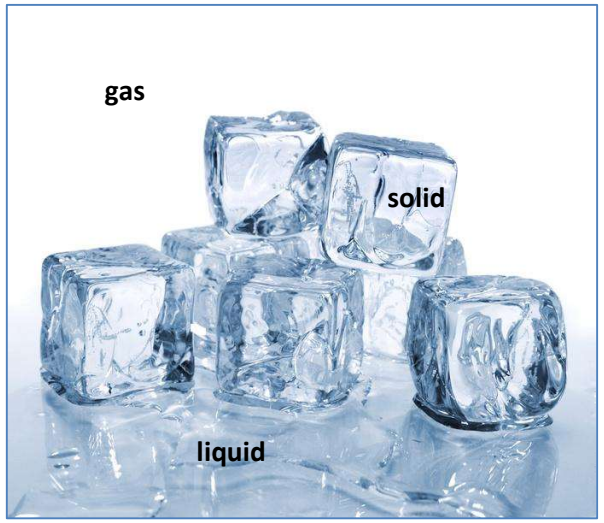
Diagram shows T vs X (composition)

Composition varies along the X axis

Pressure is constant throughout the diagram (P = 0.1 MPa = 1 bar)

THERMODYNAMICS OF MELTING

At the melting point of an element / compound / mineral, the solid and liquid phases are in equilibrium. This means that:



$$\Delta G_{melting} = 0$$

Since $\Delta G_m = \Delta H_m - T_m \Delta S_m$

then $\Delta H_m = T_m \Delta S_m$

T_m = melting temperature (in Kelvin)
ΔS_m = entropy change of melting
ΔH_m = enthalpy (heat) of melting or fusion. This can be measured using calorimetry.

Example: diopside.
We know T_m = 1391 °C, ΔH_m = 138,100 J mol^{-1}

Entropy change of melting:

$$\Delta S_m = 82.99 \text{ J mol}^{-1} \text{ K}^{-1}$$

- high, positive value as expected because melting increases the degree of disorder of the system.



Diopside

THERMODYNAMICS OF MELTING - continued

The lowest melting point of a mixture of two substances is always lower than the melting points of the individual substances.

The depression of melting point (or freezing point) is described by the equations:

$$\ln x_i = -(\Delta H_{i,m} / R)(1/T - 1/T_{i,m}) \quad \text{for phase } i$$

and

$$\ln x_j = -(\Delta H_{j,m} / R)(1/T - 1/T_{j,m}) \quad \text{for phase } j$$

} in a mixture of *i* and *j*

where x_i = fraction of *i* in the mixture
x_j = fraction of *j* in the mixture
$\Delta H_{i,m}$ = enthalpy change of melting for *i*
$\Delta H_{j,m}$ = enthalpy change of melting for *j*
R = Gas constant
T = temperature in Kelvin
$T_{i,m}$ = melting temperature of *i*
$T_{j,m}$ = melting temperature of *j*



www.crankshift.com

This applies to mixtures of substances such as antifreeze & water as well as minerals

Example: melting of diopside and anorthite

Diopside



Melting T = 1391 °C

Anorthite



http://www.wikiwand.com/en/Plagioclase

Melting T = 1553 °C

Rock containing Diopside and Anorthite



ucl.ac.uk/earth-sciences

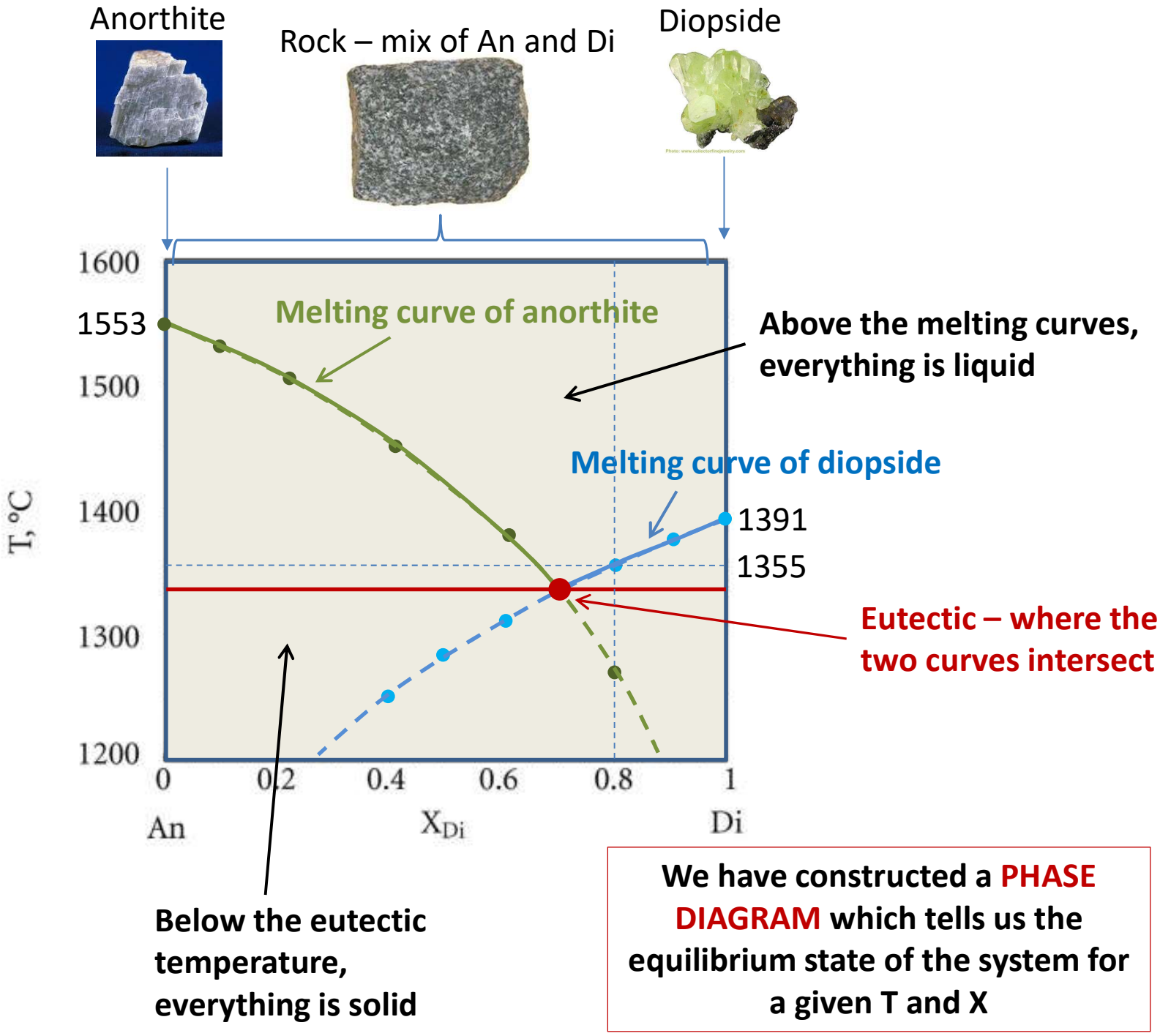
Melting T = ?? °C

For diopside, $\ln x_{Di} = -(\Delta H_{Di,m} / R)(1/T - 1/T_{Di,m})$

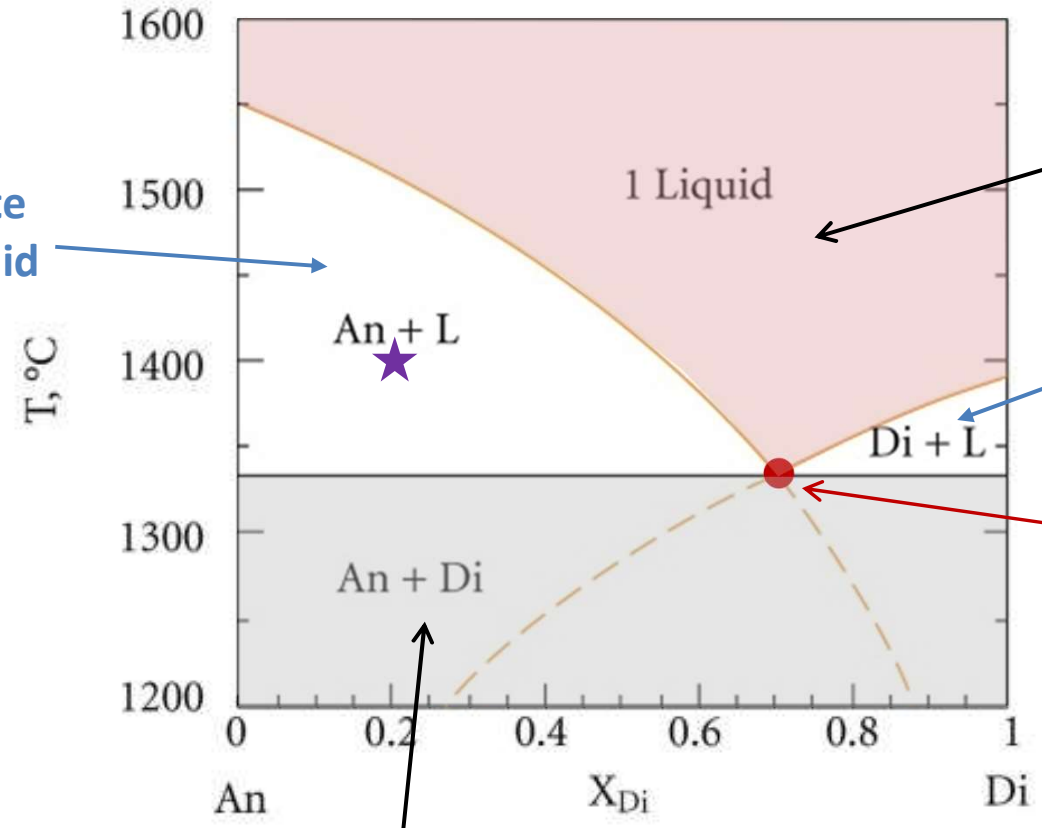
We know: $T_{Di,m}$ = 1391 °C and $\Delta H_{Di,m}$ = 138,100 J mol^{-1} *(same as 2 slides ago)*

Rearrange equation to calculate T: $T = \dfrac{\Delta H_{Di,m}}{(\Delta H_{Di,m} / T_{Di,m}) - R \ln x_{Di}}$

For a rock that is a mixture of 80% diopside and 20% anorthite, x_{Di} = 0.8
The melting temperature of this rock is given by T = 1355 °C
i.e. the rock melts completely at 1355 °C


Anorthite
Rock – mix of An and Di
Diopside
1600
1553
1500
1400
1300
1200
T, °C
Melting curve of anorthite
Above the melting curves, everything is liquid
Melting curve of diopside
1391
1355
Eutectic – where the two curves intersect
0
0.2
0.4
0.6
0.8
1
An
X_{Di}
Di
Below the eutectic temperature, everything is solid
We have constructed a PHASE DIAGRAM which tells us the equilibrium state of the system for a given T and X

Diopside – Anorthite phase diagram calculated from thermodynamic equations for melting curves



Anorthite plus liquid

Above the melting curves, everything is liquid

Diopside plus liquid

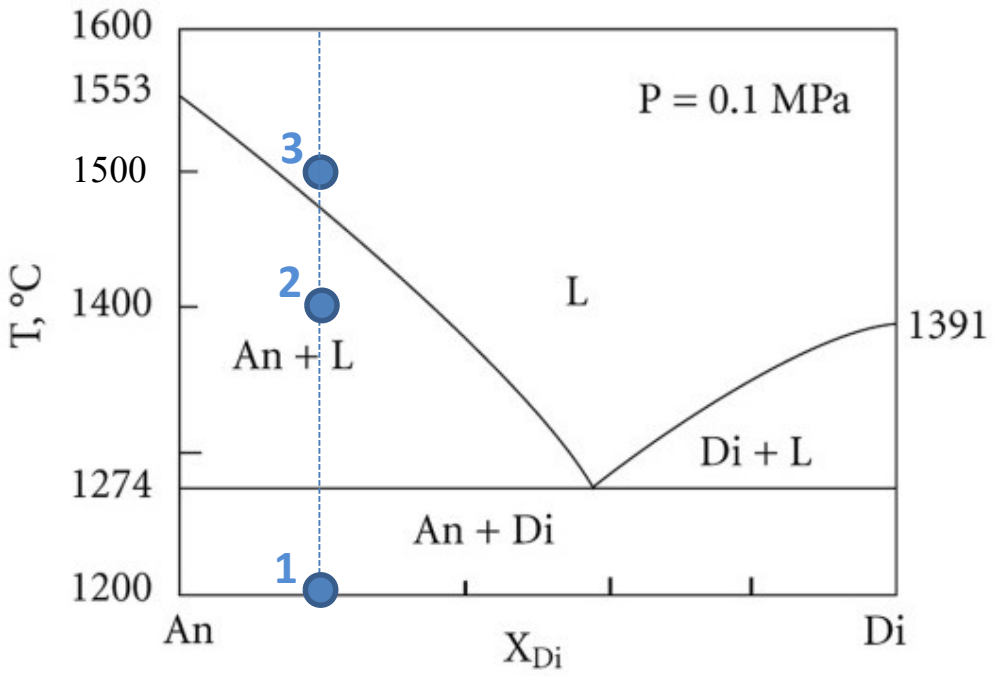
Eutectic – where the two curves intersect

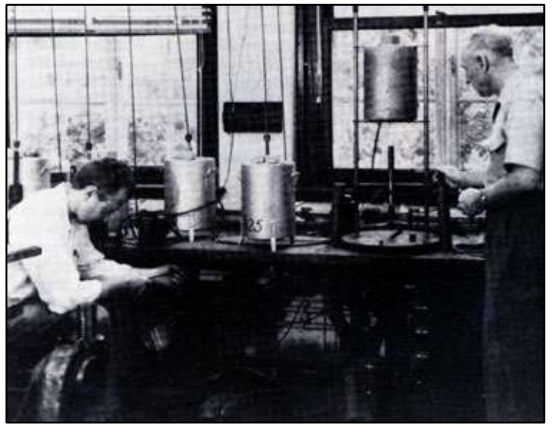
Below the eutectic temperature, everything is solid

★ Consider a mixture of 80% anorthite and 20% diopside (X_{Di} = 0.2)

★ At 1400 °C, the system consists of anorthite + liquid

Phase diagrams are also determined from experiments.





O. F. Tuttle and N. L. Bowen
https://library.gl.ciw.edu

Experiments: a mix of diopside and anorthite is heated to a known temperature, then cooled quickly to preserve the high-T reaction.

Example:

1) A mixture of 80% anorthite and 20% diopside is heated to 1200 °C. No melting occurs.
2) The mixture is heated to 1400 °C. All of the diopside and some of the anorthite melts. The experiment consists of anorthite crystals and liquid.
3) The mixture is heated to 1500 °C. The mixture is completely melted.



University of Manchester
Experimental Petrology lab

Chemistry of the Earth: Practical 6

Chemical Equilibrium (Thermodynamics)



Thermodynamics and Phase Diagrams

Practical 6:

Chemical Equilibrium: Thermodynamics and Phase Diagrams

- Familiarity with thermodynamic variables (enthalpy, entropy, Gibbs Free Energy) and the concept of equilibrium
- Construct a phase diagram from thermodynamic data
- Construct a phase diagram from experimental data
- Interpretation of basic phase diagrams and the Gibbs Phase Rule

THERMODYNAMICS IN A NUTSHELL

$PV = nRT$ — **Ideal Gas Law**

$dU = dQ - PdV$ — **First Law of Thermodynamics**

$dS \geq dQ/T$ — **Second Law of Thermodynamics**

$dU = TdS - PdV$

$dG = VdP - SdT$

$\Delta G = \Delta H - T\Delta S$

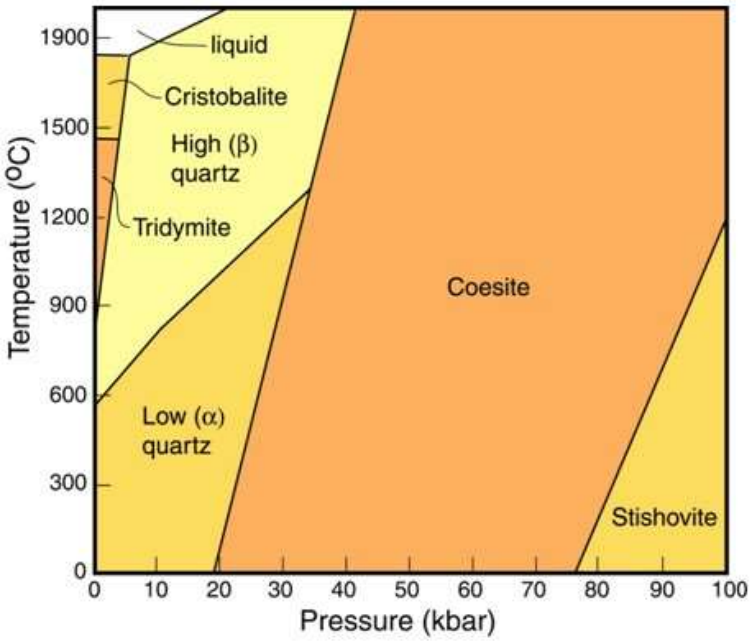
d = a very small change
Δ = a change

P = pressure
V = volume
n = number of moles
R = Gas Constant (8.314 J mol^{-1})
T = temperature **in Kelvin**
U = internal energy (sometimes written as E)
Q = heat flow from surroundings
S = entropy
H = enthalpy
G = Gibbs Free Energy

COMPONENTS:

The number of **components** in a system is the minimum number of chemical species required to describe the composition of all the phases in the system.

Phase diagram for SiO_2

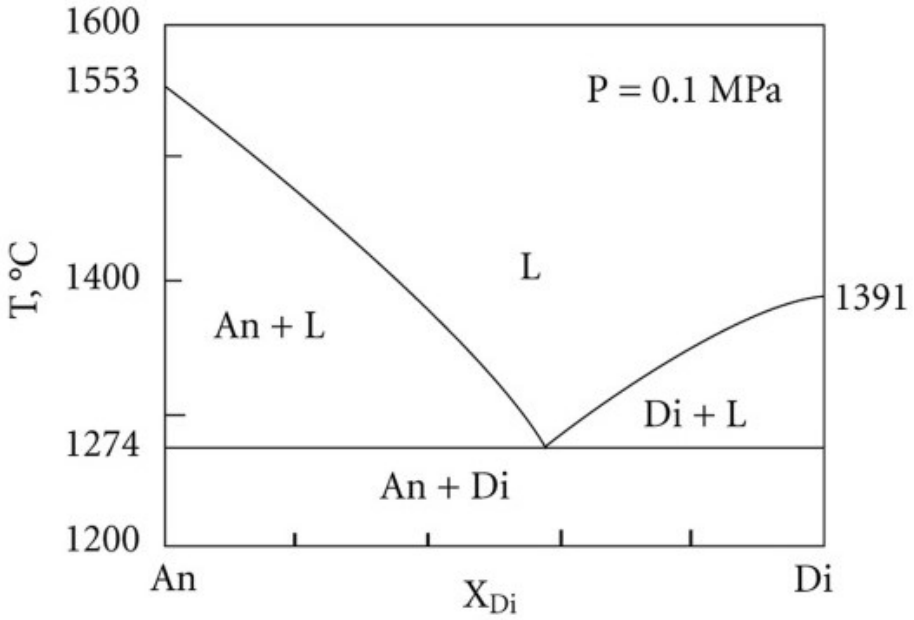


One component – SiO_2

Diagram shows T vs P

Composition (X) is constant throughout the diagram (X = SiO_2)

Phase diagram for Anorthite-Diopside



Two components – An and Di

Diagram shows T vs X (composition)

Composition varies along the X axis

Pressure is constant throughout the diagram (P = 0.1 MPa = 1 bar)

GIBBS PHASE RULE:

An equation relating the number of phases, components and thermodynamic variables in a system.

$$\mathbf{f = c - \phi + 2}$$

c = components
ϕ = phases
f = “degrees of freedom” or “variance”

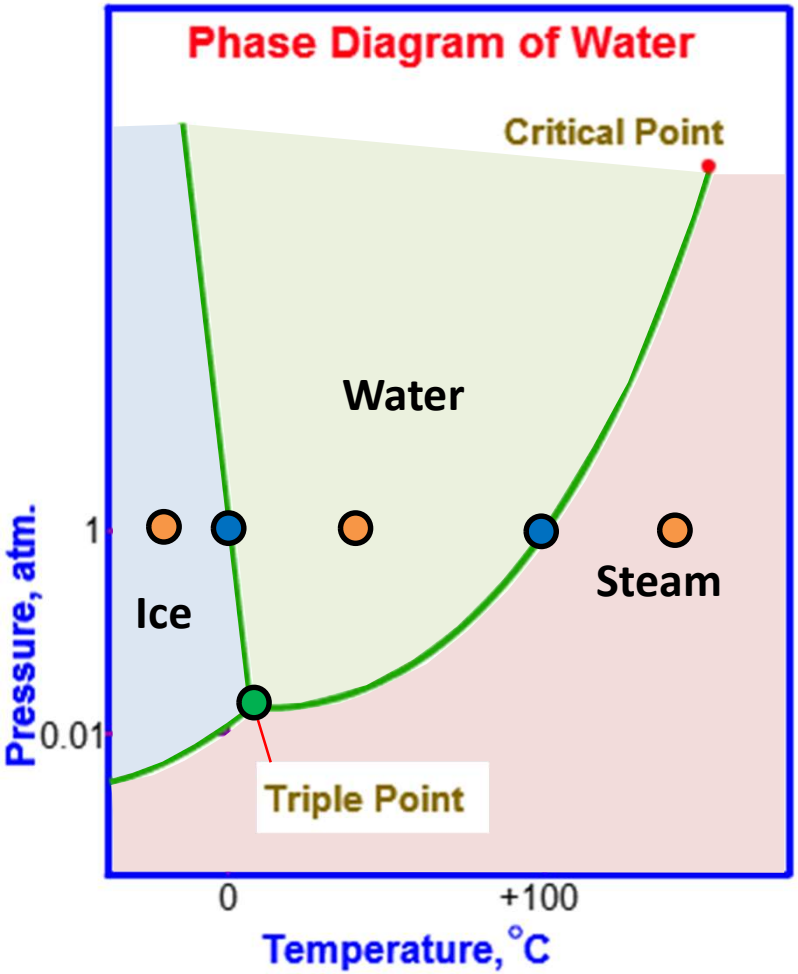
f – number of intensive variables that can be changed independently, without altering the state of the system, i.e. without changing the number of phases

f – usually consider the variables temperature (T) and pressure (P).

- The “2” in the above equation applies to a case where X is fixed, and we are considering changes in T and P (One-component phase diagram)
- If pressure is fixed e.g. in a T vs X diagram, the phase rule becomes: $\mathbf{f = c - \phi + 1}$

PHASE RULE EXAMPLE:

H_2O system



One component (H_2O)

Three phases in the diagram: ice, water, steam

Phases in equilibrium	Number of phases in equilibrium (p)	Variance (f)	Description
Ice or water or steam	1	2 (P and T)	Divariant areas
Ice ⇔ water (melting) or Water ⇔ steam (boiling) or Steam ⇔ ice (sublimation)	2	1 (P or T)	Univariant curves
Ice+water+steam	3	0 (both P and T are fixed)	Invariant point

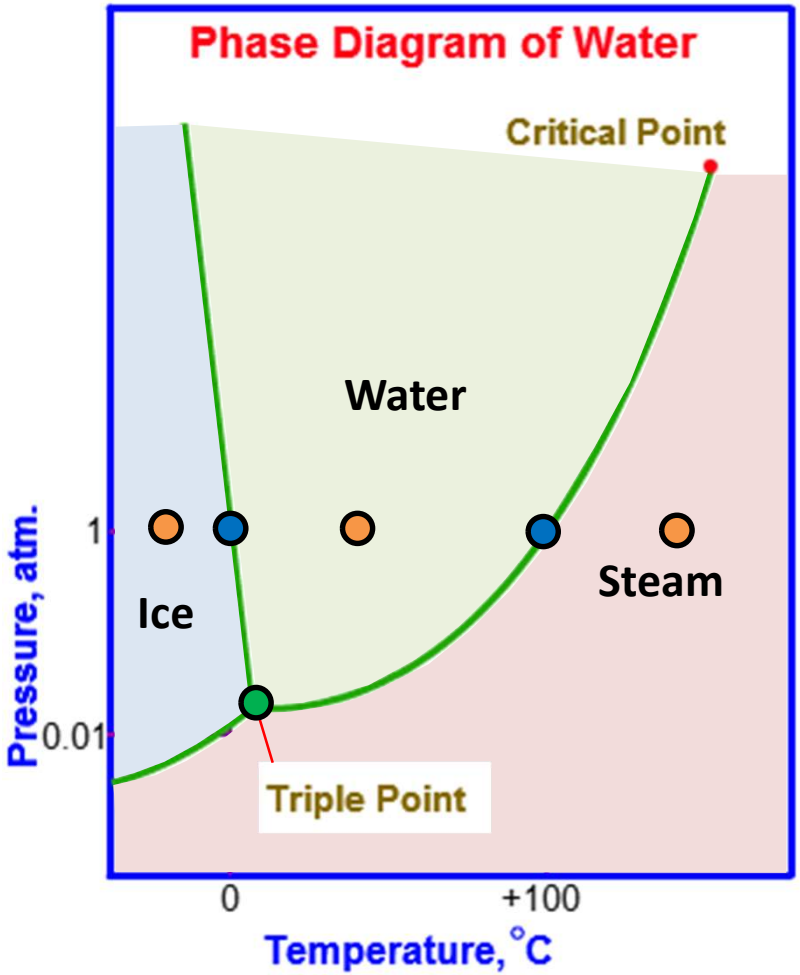
Table 4.1 Phase equilibria in the unary system H_2O (c=1) shown in Fig. 4.4.

$$f = c - \phi + 2$$

PHASE RULE EXAMPLE:

H_2O system

$$f = c - \phi + 2$$



- Where f = 2, we can make a small change in either P or T without changing the state of the system.
- Where f = 1, if we make a small change in P we must also make a change in T to keep the same state of the system (and vice versa).
- Where f = 0, we cannot change either P or T without changing the state of the system.